

Proficiency Test 2

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Task - 1

```
for(i in 1:length(mtcars)){  
  print(mtcars$mpg[1:3])  
}
```

```
[1] 21.0 21.0 22.8  
[1] 21.0 21.0 22.8  
[1] 21.0 21.0 22.8  
[1] 21.0 21.0 22.8  
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[1] 21.0 21.0 22.8  
[1] 21.0 21.0 22.8
```

Task - 2

```
count_above <- function(data,count,cutoff){ #assigns the function
  count_above

  for (i in 1:length(count)){ #defines what the function is doing
    if(count[i] > cutoff){
      count<- count+1 #for instance, the function will take the count and add 1 to it
    }
  }
  return(count) #This is what the code will give you when ran
}
```

How many flowers have a sepal length greater than 6 cm in the iris dataset?

```
count_above(iris, iris$Sepal.Length, 6)
```

```
[1] 105.1 104.9 104.7 104.6 105.0 105.4 104.6 105.0 104.4 104.9 105.4 104.8
[13] 104.8 104.3 105.8 105.7 105.4 105.1 105.7 105.1 105.4 105.1 104.6 105.1
[25] 104.8 105.0 105.0 105.2 105.2 104.7 104.8 105.4 105.2 105.5 104.9 105.0
[37] 105.5 104.9 104.4 105.1 105.0 104.5 104.4 105.0 105.1 104.8 105.1 104.6
[49] 105.3 105.0 107.0 106.4 106.9 105.5 106.5 105.7 106.3 104.9 106.6 105.2
[61] 105.0 105.9 106.0 106.1 105.6 106.7 105.6 105.8 106.2 105.6 105.9 106.1
[73] 106.3 106.1 106.4 106.6 106.8 106.7 106.0 105.7 105.5 105.5 105.8 106.0
[85] 105.4 106.0 106.7 106.3 105.6 105.5 105.5 106.1 105.8 105.0 105.6 105.7
[97] 105.7 106.2 105.1 105.7 106.3 105.8 107.1 106.3 106.5 107.6 104.9 107.3
[109] 106.7 107.2 106.5 106.4 106.8 105.7 105.8 106.4 106.5 107.7 107.7 106.0
[121] 106.9 105.6 107.7 106.3 106.7 107.2 106.2 106.1 106.4 107.2 107.4 107.9
[133] 106.4 106.3 106.1 107.7 106.3 106.4 106.0 106.9 106.7 106.9 105.8 106.8
[145] 106.7 106.7 106.3 106.5 106.2 105.9
```

How many cars have a value greater than or equal to 150 disp in the mtcars dataset?

```
count_above(mtcars, mtcars$disp, 150)
```

```
[1] 182.0 182.0 130.0 280.0 382.0 247.0 382.0 168.7 162.8 189.6 189.6 297.8
[13] 297.8 297.8 494.0 482.0 462.0 100.7 97.7 93.1 142.1 340.0 326.0 372.0
[25] 422.0 101.0 142.3 117.1 373.0 167.0 323.0 143.0
```

Task - 3

The concept of masking functions involves when a new package hides a function from an older package. For example, you install a package in R and it hides a function from an already installed package on your R.

Task - 4

```
iris$Sepal.Length[1]
```

```
[1] 5.1
```

```
iris[1,1]
```

```
[1] 5.1
```

The first code returns the first value of the iris data set for Sepal.Length.

The iris[1,1] returns the value for the first row, first column data point, which in this case is the first value of the Sepal.Length column for the iris data set. Therefore, they produce the same output.